



DEALER BEST PRACTICE SERIES

Tool for Lifting Piston and
Connecting Rod - 3500
Engines

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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The use of this tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

1. Introduction

Ferreyros has developed a tool to improve the assembly process in 3500 engines, by installing the piston and connecting rod. The tool lifts the piston and connecting rod, reducing the stress applied by the technician to lift the component, and places the piston and connecting rod directly into the cylinder block. Caterpillar does not have a tool to lift the piston and connecting rod in 3500 engines.

2. Best Practice Description

The lifting of the piston and connecting rod for installation in 3500 engines was very manual by the technician. It involved loading the piston and connecting rod from his car to block cylinder.

The safety of this procedure could be improved, and it also introduces back fatigue for the technician.



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DATE
12/7/2015

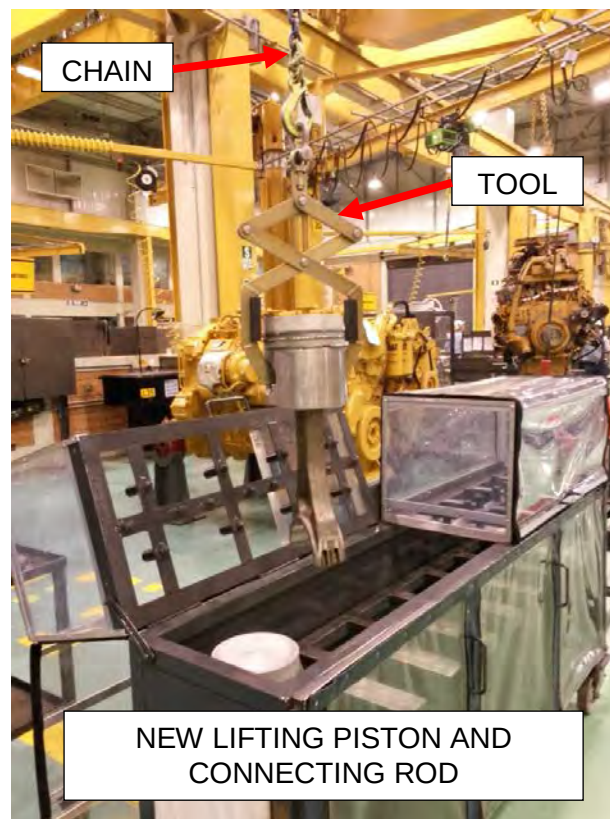
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NUMBER
1508-4.03-1388

Since tool prints are not currently available, the following photos show the fixture and process.



Proceed to install the tool on the piston and connecting rod and lift the component with a chain. The tool allows transport of the piston and the connecting rod forms safely to the cylinder block.



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The piston and connecting rod are safely installed in the cylinder block without losing its stability. This tool minimizes risks during the operation for the piston and connecting rod and the technician.



INSTALL PISTON AND CONNECTING ROD

3. Implementation Steps

- Design a tool.
- EDITOR'S NOTE: The side of the tooling will make contact with the piston rings and create potential stress on the piston ring when you move the piston / rod from the vertical into the horizontal position to install the group into the cylinder liner. Stops could be incorporated in this tooling to eliminate tooling / ring contact.
- Caterpillar also recommends developing a locking mechanism to lock grip when holding piston.

4. Benefits

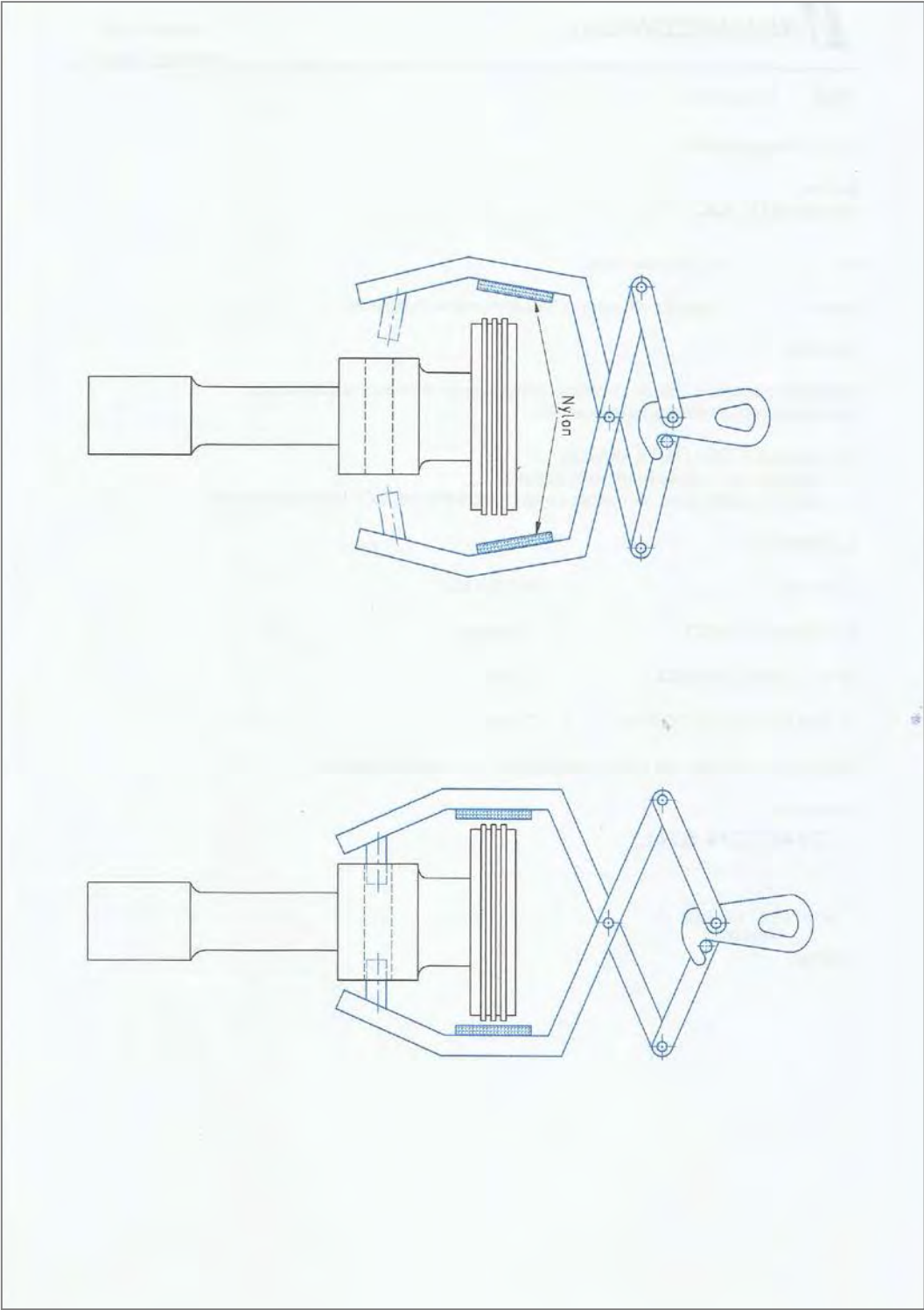
- Improve operator safety.
- Improve process efficiency.
- Prevents damage to the piston and connecting rod during install.

5. Resources Required

- Minimal expense to fabricate the tool.

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6. Supporting Attachments / References



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7. Related Best Practices

None

8. Acknowledgements

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